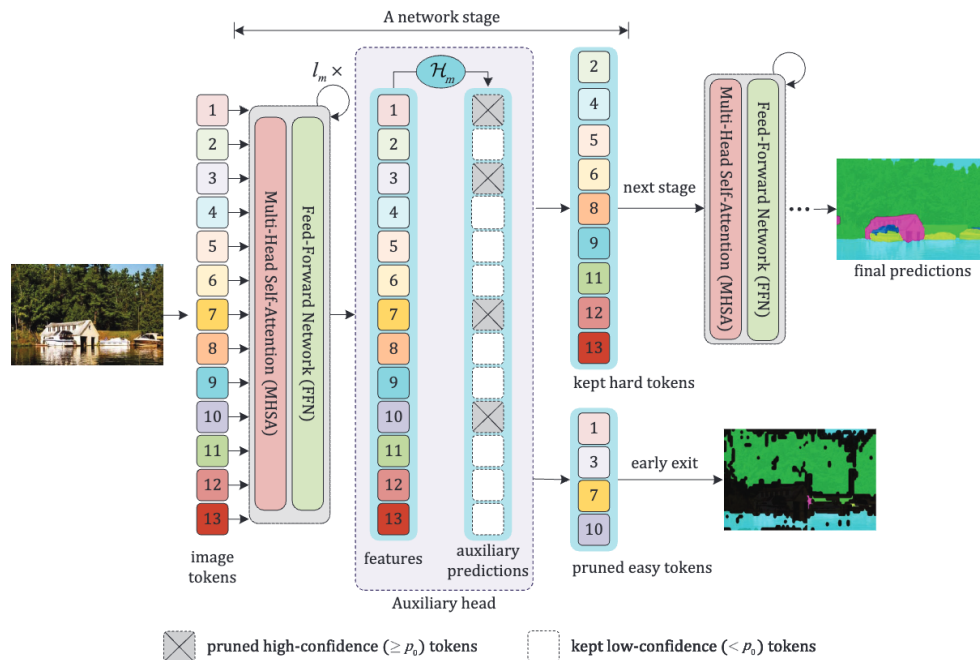


Early-exit of Transformer Network in Semantic Segmentation



Description: This project will explore the theory behind early exit when applied singularly to tokens in a Transformer network. Specifically, we want to speed up the encoder in the transformer architecture by dynamically recognizing the less important tokens (like the ones representing sky or grass) and having them sent to the decoder head without further middle computations. This will save important computations avoiding using unnecessary tokens in the quadratic computations of the attention component.

Keywords: Vision Transformer, Dynamic neural networks, Early Exit models

Resources:

<https://arxiv.org/pdf/2501.07451v1>

https://openaccess.thecvf.com/content/ICCV2023/papers/Tang_Dynamic_Token_Pruning_in_Plain_Vision_Transformers_for_Semantic_Segmentation_ICCV_2023_paper.pdf

Required level:

Master student with experience in computer vision, deep learning, and Pytorch. You should have trained deep learning model before.

Contact: fabmo@dtu.dk